Impact of Model Complexity on Transfer Learning-Based Pneumonia Detection Using Public Chest X-Ray Images

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Abstract*—This paper examines how architectural complexity influences the effectiveness of transfer learning models for automated pneumonia detection from chest X-ray images. Three convolutional neural network (CNN) backbones of differing complexity, MobileNetV2, ResNet50 and EfficientNetB0, were evaluated within a single standardised transfer learning pipeline. Each backbone was pre-trained on ImageNet, fitted with an identical classification head, and trained in two phases (frozen-base warm-up followed by selective fine-tuning) with class weighting to mitigate class imbalance and early stopping to ensure convergence. Beyond predictive accuracy, the study reports computational efficiency in terms of parameter count, floating-point operations (FLOPs) and inference latency, together with bootstrap confidence intervals for statistical validation. The experimental findings show that, once preprocessing is matched to each backbone and the models are trained to convergence, architectural complexity does not yield large differences in discriminative accuracy: all three models achieve a comparable ROC-AUC of between 0.937 and 0.962. The lightweight MobileNetV2 attains the highest ROC-AUC (0.962) and the lowest computational cost, whereas EfficientNetB0 achieves the highest test accuracy (87.0%) and the most balanced sensitivity–specificity profile. The principal differences between models therefore emerge not in overall accuracy but in the operating-point trade-off and computational footprint, which is the property most relevant to deployment on small, imbalanced medical imaging datasets.

***Index Terms*—Pneumonia Detection, Transfer Learning, Deep Learning, Medical Imaging, Model Complexity, Chest X-Ray**

I. Introduction

Pneumonia remains a major global health burden and a considerable cause of morbidity and mortality across both paediatric and adult populations. Timely and precise diagnosis is a key factor in ensuring appropriate clinical intervention and in limiting adverse outcomes. Chest radiography is among the most common diagnostic modalities for pneumonia; however, manual interpretation of chest X-ray images is time-consuming and prone to inter-observer variability.

Recent progress in deep learning has shown promising results in automated medical image analysis tasks [18]. In particular, convolutional neural networks (CNNs) have demonstrated a strong capability for extracting hierarchical image features for disease classification [1], [5]. Transfer learning, in which models trained on large-scale datasets such as ImageNet are adapted to a medical imaging task, has become a widely adopted approach to pneumonia detection because annotated medical datasets are frequently unavailable in large quantities [7], [8].

Although numerous studies report high performance using deep CNN architectures [3], [10], the relationship between model complexity and generalisation performance in pneumonia detection remains insufficiently examined. Larger architectures with more parameters do not always outperform smaller ones, particularly on imbalanced or small datasets [2], [15]. A recurring difficulty in the existing literature is that comparisons across architectures are often confounded by inconsistent preprocessing, differing training budgets and the absence of computational-cost reporting, which makes it hard to attribute observed differences to architecture rather than to experimental artefacts.

This paper addresses that gap by providing a controlled comparison of three CNN backbones of differing complexity under a single, carefully matched experimental protocol. The specific contributions of this work are as follows. First, we establish a standardised pipeline in which each backbone receives its own architecture-specific preprocessing, removing a common confound in cross-architecture comparison. Second, we adopt a two-phase training regime with selective fine-tuning and early stopping, so that each model is trained to convergence rather than for an arbitrary fixed number of epochs. Third, we report computational efficiency, namely parameter count, FLOPs and inference latency, alongside predictive metrics, and we provide bootstrap confidence intervals so that differences between models can be assessed for statistical significance. These design choices distinguish the present study from prior work that compares architectures without controlling for preprocessing, training budget or computational cost.

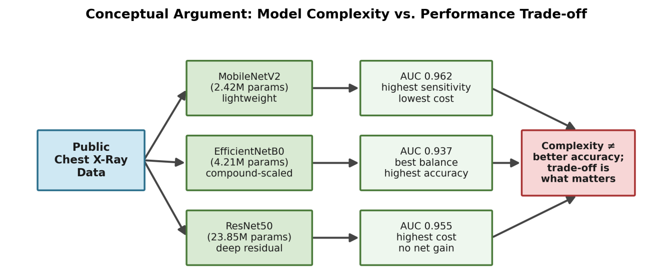

Fig. 1. Conceptual overview: chest X-ray images are passed through transfer learning backbones of increasing complexity, and the resulting predictive performance and computational cost are compared.

II. Related Work

Research on deep learning methods for pneumonia detection has expanded rapidly in recent years. Comprehensive surveys highlight the evolution of CNN-based architectures and transfer learning strategies for chest X-ray analysis [1], [9]. Several studies report strong performance from pre-trained models fine-tuned on pneumonia datasets.

Transfer learning has been widely adopted owing to the limited size of medical imaging datasets [7], [8]. Salehi et al. [3] demonstrated effective pneumonia classification using deep transfer learning on paediatric chest X-rays, while Manickam et al. [10] compared multiple pre-trained architectures for automated detection. Ensemble approaches have also been proposed to improve robustness [6]. More recent work has explored architectural variations, attention mechanisms and hyperparameter customisation to enhance performance [12], [16]. However, several studies note that model performance can degrade under class imbalance, and that imbalance-mitigation strategies are often necessary [2], [15].

Pre-trained backbones such as VGGNet, ResNet, MobileNet and EfficientNet allow models to reuse visual representations learned from large-scale datasets such as ImageNet, improving training efficiency and generalisation [7]. Some studies suggest that lightweight architectures such as MobileNet can offer competitive performance while maintaining a lower computational cost [14], whereas others emphasise that imbalanced training distributions can significantly affect model reliability and diagnostic sensitivity [2], [15].

In spite of these developments, there has been little systematic experimentation that isolates the effect of architectural complexity on generalisation within a standardised experimental setting that also controls preprocessing and reports computational cost. The present paper provides such a controlled comparison between lightweight and deeper transfer learning models, with the aim of better understanding their behaviour on the pneumonia detection task.

III. Methodology

A. Conceptual Framework

The proposed pneumonia detection system links dataset preparation, transfer learning, model training and performance assessment. The pipeline begins with the publicly available chest X-ray images, which are preprocessed and augmented to improve generalisation during training. Pre-trained CNN backbones are then adapted by transfer learning, trained under a standardised configuration, and evaluated using accuracy, sensitivity, specificity and ROC-AUC, together with computational-cost metrics. The overall workflow is illustrated in Fig. 2.

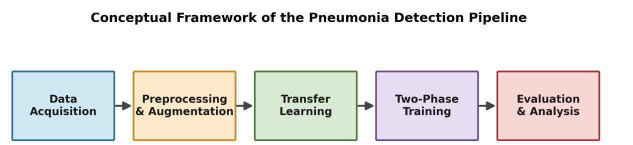

Fig. 2. Conceptual framework of the proposed transfer learning-based pneumonia detection pipeline.

B. Dataset Description

The publicly available Chest X-Ray Images (Pneumonia) dataset, obtained from Kaggle [8], was used in this study. The dataset contains anterior–posterior chest radiographs categorised into two classes, *Normal* and *Pneumonia*, and is widely used in pneumonia detection research owing to its accessibility and reproducibility [1]. The established training and test divisions were retained. The original training set was further divided into training (85%) and validation (15%) subsets using an internal validation split to ensure consistent comparison across models. The training set contained 5,216 images (1,341 Normal and 3,875 Pneumonia), confirming a substantial class imbalance of approximately 1:2.9 in favour of the Pneumonia class. The independent test set comprised 624 images (234 Normal and 390 Pneumonia). The overall experimental framework is illustrated in Fig. 3.

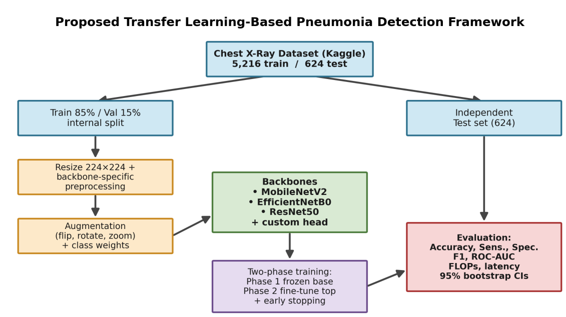

Fig. 3. Proposed transfer learning-based pneumonia detection framework.

C. Preprocessing and Data Augmentation

All images were resized to 224×224 pixels to match the input requirements of the pre-trained architectures. To ensure a fair comparison, each backbone was supplied with the inputs it was originally pre-trained on by applying its own official preprocessing function rather than a single shared rescaling rule. Specifically, MobileNetV2 inputs were scaled to the range $[-1, 1]$, ResNet50 inputs used channel-wise mean subtraction in the Caffe convention, and EfficientNetB0 inputs used its built-in normalisation. This per-backbone preprocessing is important because applying a uniform $[0, 1]$ rescaling to all models, as is sometimes done, places ResNet50 and EfficientNetB0 outside their expected input distribution and substantially degrades their performance. To reduce overfitting and improve generalisation, the training data were augmented using horizontal flipping, small rotations (up to approximately ten degrees) and zoom transformations. The validation and test sets were not augmented.

D. Transfer Learning Architectures

Three pre-trained CNN backbones, spanning a wide range of parameter counts, were evaluated: MobileNetV2 (lightweight), EfficientNetB0 (compound-scaled) and ResNet50 (deep residual). For each backbone, the ImageNet pre-trained weights were used for initialisation, the original classification head was removed, and a common custom head was attached, consisting of a global average pooling layer, a fully connected dense layer of 128 units with ReLU activation, a dropout layer with rate 0.3, and a single sigmoid output unit for binary classification. Using an identical head across all backbones ensures that observed differences arise from the backbone rather than from the classifier. The generic model design is illustrated in Fig. 4.

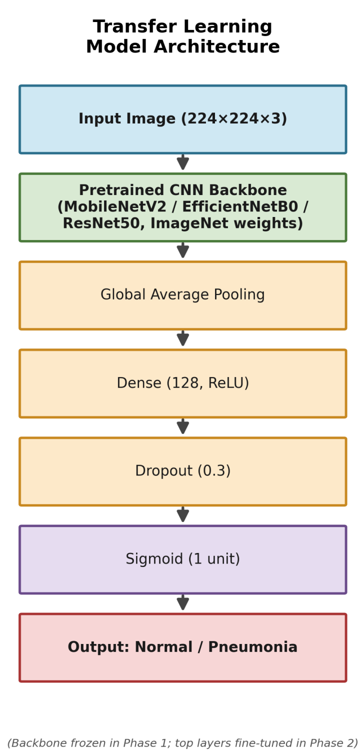

Fig. 4. Transfer learning model architecture used for pneumonia detection.

E. Training Configuration

Each model was trained in two phases using the Adam optimiser and a binary cross-entropy loss. In the first phase, the convolutional base was frozen and only the custom head was trained, with a learning rate of 1×10^{-3}. In the second phase, the top thirty layers of the backbone were unfrozen and the model was fine-tuned with a reduced learning rate of 1×10^{-5} to adapt the higher-level features to the chest X-ray domain while preserving the lower-level ImageNet representations. Rather than training for a fixed number of epochs, each phase used early stopping (monitoring validation ROC-AUC, patience of five epochs, with restoration of the best weights) and a learning-rate-reduction-on-plateau schedule, with an upper bound of 25 epochs for the warm-up phase and 15 for the fine-tuning phase. This regime allows deeper models the additional training iterations they require to converge while preventing overfitting on the smaller dataset. To address the class imbalance described above, class weights inversely proportional to class frequency were applied to all three models during training, ensuring that the comparison was consistent across architectures. A batch size of 32 was used throughout. All experiments were conducted in Google Colab. Parameter counts and training times were recorded for the computational-complexity analysis.

F. Proposed Algorithm

Algorithm 1 Transfer Learning-Based Pneumonia Detection Framework

```
Load chest X-ray dataset
Split training data into training (85%) and validation (15%); retain the supplied test set
Resize images to 224 × 224 pixels
Compute class weights from the training-class frequencies
for all backbones in {MobileNetV2, ResNet50, EfficientNetB0} do
    Apply backbone-specific preprocessing and training augmentation
    Initialise backbone with ImageNet weights; attach custom binary head
    Phase 1: freeze base; train head with learning rate 1 × 10^-3 and class weights until early stopping
    Phase 2: unfreeze top layers; fine-tune with learning rate 1 × 10^-5 and class weights until early stopping
    Evaluate on the independent test set
    Record parameters, FLOPs and inference latency
end for
Report Accuracy, Sensitivity, Specificity, ROC-AUC and bootstrap confidence intervals
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G. Evaluation Metrics

The independent test set was assessed using accuracy, precision, recall (sensitivity), specificity, F1-score, the area under the receiver operating characteristic curve (ROC-AUC) and the confusion matrix. Sensitivity corresponds to pneumonia recall, while specificity is the true-negative rate on the Normal class. High recall is particularly important in pneumonia detection because of the clinical cost of false negatives [4], [5]. To assess whether differences between models were statistically meaningful, 95% confidence intervals for accuracy and ROC-AUC were estimated by bootstrap resampling of the test set over 1,000 replicates. In addition, computational efficiency was quantified using the number of parameters, the number of floating-point operations (FLOPs) and the mean single-image inference latency.

IV. Results and Discussion

The comparative performance of the evaluated architectures is summarised in Table I, and a detailed classification report is given in Table II. The confusion matrices and the combined ROC curves are shown in Figs. 5 and 6, respectively.

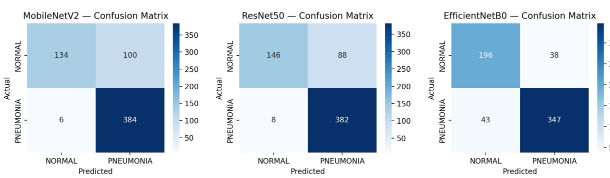

Fig. 5. Confusion matrices of the evaluated models on the test set, showing the distribution of true negatives (TN), false positives (FP), false negatives (FN) and true positives (TP) for MobileNetV2, ResNet50 and EfficientNetB0.

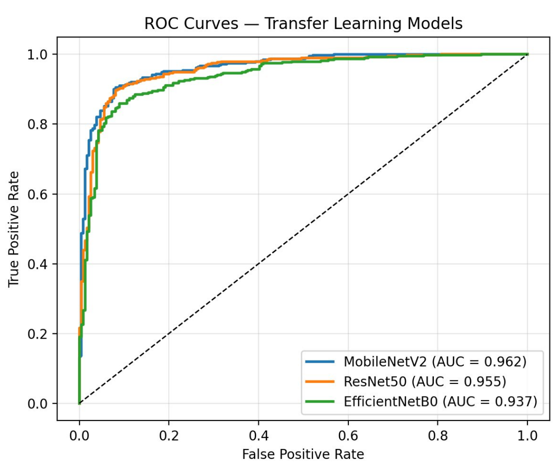

Fig. 6. Receiver operating characteristic (ROC) curves for the three transfer learning models on the independent test set.

TABLE I
Comparative Performance and Computational Cost of Transfer Learning Models

Model	Params	GFLOPs	Latency (ms)	Test Acc	ROC-AUC
MobileNetV2	2.42M	0.61	297.4	83.0%	0.962
ResNet50	23.85M	7.75	602.6	84.6%	0.955
EfficientNetB0	4.21M	0.80	495.6	87.0%	0.937

TABLE II
Classification Performance of the Evaluated Models (with 95% bootstrap confidence intervals for accuracy and ROC-AUC)

Model	Sens.	Spec.	Prec.	F1	Acc 95% CI	AUC 95% CI
MobileNetV2	0.985	0.573	0.793	0.879	[0.803, 0.861]	[0.948, 0.975]
ResNet50	0.979	0.624	0.813	0.888	[0.817, 0.873]	[0.938, 0.970]
EfficientNetB0	0.890	0.838	0.901	0.895	[0.843, 0.894]	[0.918, 0.955]

A. Predictive Performance

Once preprocessing was matched to each backbone and the models were trained to convergence with class weighting, the three architectures achieved broadly comparable discriminative performance, with ROC-AUC values ranging from 0.937 to 0.962. The overlapping 95% confidence intervals for ROC-AUC indicate that the differences between models in overall discriminative ability are small and not strongly separated statistically. This is a notable contrast with comparisons in which preprocessing is not matched to the backbone, where deeper models can appear to fail; the present results suggest that such failures are largely an artefact of input handling rather than of architecture.

The more informative differences appear in the operating-point behaviour. MobileNetV2 achieved the highest ROC-AUC (0.962) and the highest sensitivity (0.985), but at a comparatively low specificity (0.573), meaning that it correctly identifies almost all pneumonia cases while misclassifying a substantial fraction of normal cases. ResNet50 produced a similar profile (sensitivity 0.979, specificity 0.624) despite having roughly ten times as many parameters. EfficientNetB0 achieved the highest test accuracy (87.0%), the highest precision (0.901) and by far the most balanced sensitivity–specificity trade-off (0.890 and 0.838 respectively), giving the strongest F1-score (0.895).

B. Clinical Interpretation

Clinically, high recall is the priority in pneumonia detection because false negatives carry the greatest risk; however, an excessively low specificity, as observed for MobileNetV2 and ResNet50, leads to many false alarms and unnecessary follow-up. EfficientNetB0's more balanced profile may therefore be preferable where the cost of false positives is non-trivial, whereas the high-sensitivity behaviour of MobileNetV2 may be acceptable in a triage setting where missed cases are the dominant concern. The appropriate choice is thus governed less by overall accuracy than by the desired operating point.

C. Computational Complexity Analysis

The three models differ markedly in computational cost. MobileNetV2 is the most efficient by a wide margin, with only 2.42 million parameters, 0.61 GFLOPs and the lowest inference latency (297 ms), yet it attains the highest ROC-AUC. ResNet50, with 23.85 million parameters and 7.75 GFLOPs, is roughly an order of magnitude more expensive in both parameters and FLOPs and has the highest latency (603 ms), without a corresponding gain in discriminative performance. EfficientNetB0 occupies an intermediate position (4.21 million parameters, 0.80 GFLOPs) while delivering the most balanced predictive profile. Taken together, these results indicate that increased architectural depth and parameter count do not translate into better generalisation on this small, imbalanced dataset, and that lightweight or compound-scaled architectures offer a more favourable trade-off between computational cost and predictive performance.

D. Summary of Findings

The central finding is that, under a controlled and fairly matched protocol, model complexity is not the dominant determinant of accuracy for this task; all three backbones are competitive once trained appropriately. The meaningful distinctions lie in the sensitivity–specificity trade-off and in computational cost. MobileNetV2 is the strongest choice when efficiency and sensitivity are paramount, while EfficientNetB0 provides the most balanced and accurate operating point at a modest computational cost. This refines, rather than simply confirms, the common claim that "smaller is better": smaller models are highly competitive, but the better practical recommendation depends on the clinical operating point and resource constraints.

V. Conclusion

This paper investigated the effect of model complexity on transfer learning-based pneumonia detection using a publicly available chest X-ray dataset. Three pre-trained CNN backbones, MobileNetV2, ResNet50 and EfficientNetB0, were compared within a single standardised pipeline that matched preprocessing to each backbone, trained every model to convergence using a two-phase frozen-then-fine-tuned regime with class weighting, and reported computational cost and bootstrap confidence intervals alongside predictive metrics.

The key contribution of the work is methodological as well as empirical: by removing the preprocessing and training-budget confounds that affect many cross-architecture comparisons, the study shows that the three architectures achieve comparable discriminative performance (ROC-AUC 0.937–0.962) and that the practically important differences lie in the sensitivity–specificity trade-off and computational footprint. MobileNetV2 offered the best efficiency and the highest sensitivity, ResNet50 offered no advantage commensurate with its far greater cost, and EfficientNetB0 provided the most balanced and accurate operating point. These findings indicate that, for small and imbalanced medical imaging datasets, careful experimental control and operating-point selection matter more than raw architectural complexity.

Limitations and Future Work

A single publicly available dataset was used, which may not capture the clinical variability of real-world settings; cross-dataset and multi-institutional validation is needed to assess broader generalisability. The fine-tuning was limited to the top layers of each backbone, and a more extensive fine-tuning study, together with stronger imbalance-mitigation strategies and hybrid or transformer-based architectures, would provide a more comprehensive comparison. Future work will therefore extend the evaluation to additional architectures and datasets, and will examine calibration and decision-threshold selection so that the reported operating-point trade-offs can be tuned to specific clinical requirements.

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